Chapter 8, Problem 6RQ

Problem

A parallel *RLC* circuit has $L = 2$ H and $C = 0.25$ F. The value of R that will produce a unity neper frequency is:

(a) 0.5 Ω

(b) 1 Ω

(c) 2 Ω

(d) 4 Ω

Step-by-step solution

Step 1 of 2

For a parallel *RLC* circuit,

Consider the expression for Neper frequency.

$$\alpha = \frac{1}{2RC} \quad \ldots\ldots (1)$$

Step 2 of 2

Substitute $1\ \text{Np/s}$ for α and $0.25\ \text{F}$ for C.

$$1 = \frac{1}{2R(0.25)}$$

$$\begin{aligned} R &= \frac{1}{0.5} \\ &= 2\ \Omega \end{aligned}$$

Therefore, the required value of the resistance is $\boxed{2\ \Omega}$.